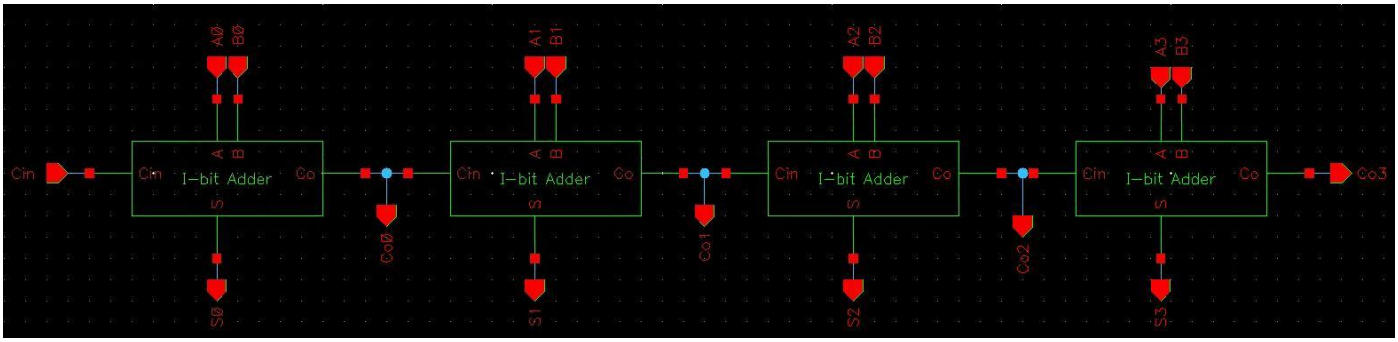
CUSTOM DIGITAL IC DESIGN

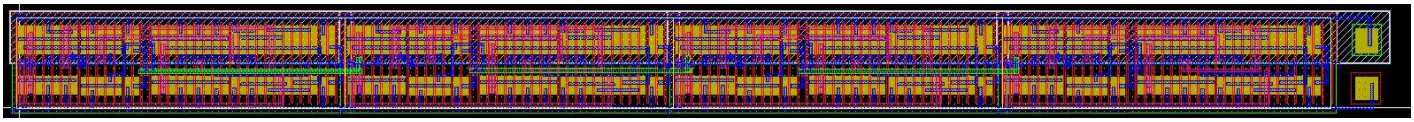
32 nm four-bit CMOS adder

Ruthvik Reddy Anthireddigari / Arizona State University / Fall 2025

I increased transistor widths on the carry path and output stages while keeping lower-bit stages at minimum size. Post-layout simulation gave 277 ps worst-case delay and 47.3 µW average power.

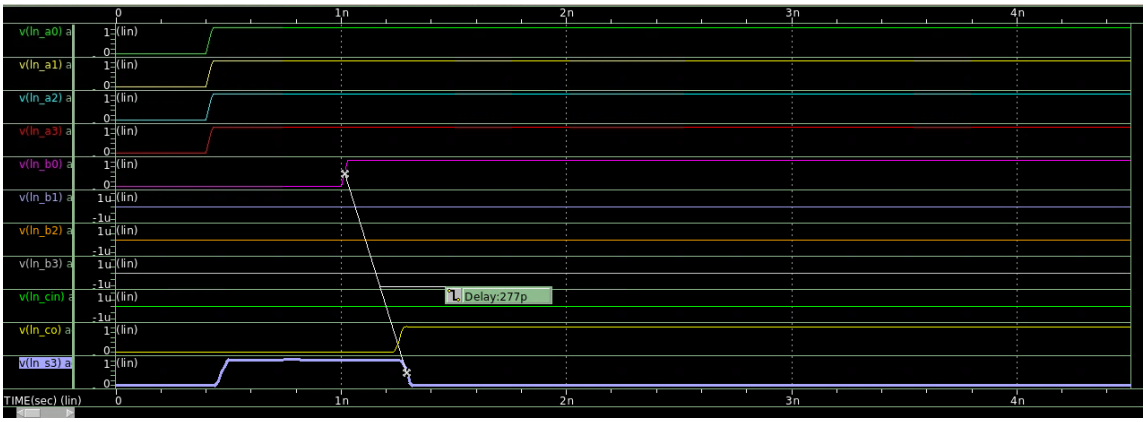
Four-bit schematic and layout





The layout occupied 40.083 µm². Carry-propagate and carry-out stages used roughly 2-3× sizing, with a smaller increase for the most significant sum output.

Extracted timing



Worst-case delay increased from 152 ps before layout to 277 ps after extraction.